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Investigating the Relationship Between Bus Network Topology and Temporal Ridership Patterns: A Case Study in Singapore

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Abstract

To advance the understanding of passenger temporal ridership profiles in urban bus systems, this study examines the relationship between network topological characteristics and temporal ridership patterns. Using Singapore's bus system as a case study, stop-level hourly passenger profiles are constructed, and a clustering approach is applied to identify five typical temporal patterns: mixed-use central business district (CBD), western job hubs, peripheral residential regions, near-central residential regions, and school-oriented zones. Adopting the complex network theory, the bus network is represented in both L-space and P-space, allowing to compute key topological descriptors and to examine degree distributions and small-world features. A multinomial logit (MNL) model is employed to investigate the relationships between temporal ridership clusters and network topology. The results confirm significant associations and identify that residential origins exhibit stronger outward connectivity and bridging roles, whereas job and school destinations demonstrate concentrated inflows and high local transitivity within dense route cliques. Additionally, incorporating network topology significantly improves model performance, highlighting that network context provides additional explanatory power for temporal ridership patterns beyond the urban environment alone. By linking network topology with temporal ridership dynamics, this study provides a network-based perspective that can inform more efficient and adaptive public bus network planning.

Keywords: urban bus system, temporal ridership pattern, clustering analysis, network topology, multinomial logit model

1 Introduction

Understanding temporal ridership patterns in public bus systems is essential for informing service planning and operational adjustments. Existing studies on transport node-level ridership variability have largely emphasized external determinants, such as the population and employment densities, land use composition, and multimodal accessibility (El Mahrsi et al. 2017; Gan et al. 2020; Li et al. 2022; Liu et al. 2019; Wang et al. 2022; Zhao et al. 2019). However, the internal topological properties of transit networks, i.e., the connectivity and relational context among stops, have received comparatively limited attention as explanatory factors for temporal demand.

Complex network theory provides a robust analytical framework for quantifying such topological characteristics using indicators such as degree, betweenness, clustering coefficient, and average shortest path length. These metrics have been widely applied to assess transport network efficiency, robustness, and vulnerability (Gecchele et al. 2019; He et al. 2025; Luo et al. 2020; Wang et al. 2024), yet their relationships with temporal ridership patterns remain underexplored. Transport networks often exhibit heavy-tailed degree distributions indicative of hub dominance, small-world properties characterized by short path lengths and high local clustering, and hierarchical centrality structures implying accessibility gradients and bridging roles (Luo et al. 2020). These features can plausibly influence passenger flows by shaping route choice, transfer opportunities, and the spatial concentration of activity. Although emerging studies link centrality measures to aggregate ridership volumes (Luo et al. 2020), systematic investigation of how network topology relates to time-varying ridership profiles is still scarce.

This study addresses this gap by jointly analyzing the temporal patterns of stop-level passenger volumes and the network topology of a large-scale urban bus system, and by quantitatively examining their relationships. Using the

Singapore bus network as a case study, the analysis framework proceeds in three phases. First, hourly boarding time series are constructed for each bus stop, and a clustering approach is applied to identify typical temporal patterns. Second, the bus network is represented in both L-space and P-space, and its topological characteristics are quantified and analyzed. Third, the multinomial logit (MNL) model is employed to examine the relationships between the identified temporal clusters and topological features. This purely topological model is compared with specifications incorporating land use features to assess the complementarity of topological features.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 describes the methodological framework, including the clustering approach, network topological measures, and the MNL model. Section 4 introduces the case and describes the data collection and preprocessing procedures. Section 5 examines the temporal ridership patterns, network topological features, and discusses their relationships. Section 6 concludes the study and outlines directions for future research.

2 Literature Review

2.1 Ridership temporal patterns of transit passenger volumes

Numerous studies have examined time-of-day ridership patterns using large-scale trip data to gain deeper insights into the dynamic variations of travel demand (El Mahrsi et al. 2017; Gan et al. 2020; Kim and Jang 2022; Liu et al. 2019; Zhao et al. 2019). Based on similarities in waveform profiles, clustering methods are commonly applied to categorize these patterns into distinct groups. For example, Zhao et al. (2019) et Gan et al. (2020) analyzed smart card data from the Nanjing metro system, extracting hourly boarding and alighting flows and employing clustering analysis to infer station functions. Kim and Jang (2022) examined subway ridership sequences in Seoul and applied hierarchical clustering to classify stations into four groups. Kim (2018) compared ridership patterns between subway and taxi trips in Seoul.

A common finding across these studies is the existence of two dominant patterns of boarding ridership: morning peak-dominant stations (typically 7:00-9:00) and afternoon peak-dominant stations (typically 17:00-19:00) (Gan et al. 2020; Kim and Jang 2022; Zhao et al. 2019). Specifically, pronounced morning boarding peaks and afternoon alighting peaks are closely associated with residential areas, whereas morning alighting and afternoon boarding peaks are linked to employment-related zones. Gan et al. (2020) defined these two types of stations as residential-oriented and employment-oriented, respectively, and demonstrated that they account for roughly 36% of stations in an urban metro system. Moreover, their MNL analysis revealed a statistically significant relationship between ridership clusters and land use. Similarly, Choi et al. (2022) found significant associations between taxi ridership patterns and local land use. However, existing research has primarily explained ridership patterns using external urban environmental features such as land use and socioeconomic factors, while the influence of internal network topological structures has received limited attention.

2.2 Transport network topology analysis

The complex network theory conceptualizes a transport system as a graph consisting of two primary components: nodes and edges. Depending on how the two elements are defined, two typical representations are commonly used to model transit networks: L-space and P-space. (von Ferber et al. 2009; Shanmukhappa et al. 2019; Taylor 2017). In L-space, transit stops or stations are represented as nodes, and route segments connecting consecutive stations are modeled as edges. This intuitive representation is close to the physical layout of the network and is widely applied (Abdelaty et al. 2020; de Regt et al. 2019; Sun et al. 2015). The path length between nodes reflects either the travel distance (for weighted networks) or the number of stations traversed (for unweighted networks). In P-space, edges are created between all stops served by the same transit line, thereby capturing transfer-free connections. Path length reflects the number of transfers, making P-space networks substantially denser than L-space networks. For example, Meng et al. (2020) found that the Shenzhen metro network contained approximately 13 times more edges in P-space than in L-space. P-space thus extends the applicability of complex network theory to a broader range of transport systems, including less tangible networks such as airports (Lordan et al. 2014), maritime shipping routes (Tocchi et al. 2022), bike-sharing origin-destinations (ODs) (Wu and Kim 2020), and taxi ODs (Liu et al. 2015).

Complex network theory provides a framework for quantifying the topological characteristics of transport networks based on the graphical arrangement and connectivity (Taylor 2017), using indicators such as degree, closeness, betweenness, clustering, and average shortest path length. Degree evaluates the importance of a node according to its direct neighbors, and is often interpreted as the connectivity of a transport station (Lordan et al. 2014; Wu and Kim 2020). Betweenness and closeness capture a node's importance in terms of its shortest path connections to all other nodes, reflecting its role in bridging flows and its overall accessibility. Clustering measures the local cohesion of a node and its neighbors, indicating the extent to which neighbors are directly connected without passing through the focal node. Additionally, the transport network often presents scale-free and small-world properties (Lin and Ban 2013; Meng et al. 2020; Sienkiewicz and Hołyst 2005). A scale-free network is characterized by a degree distribution that follows a power law (Barabási and Pósfai 2016), in which a minority of nodes hold disproportionately high degrees and dominate the network. In real-world transport systems, such nodes typically correspond to major hubs, which are both critical and vulnerable and therefore warrant particular attention in network planning and resilience studies. The small-world property, meanwhile, reflects a network structure with short average path lengths and high clustering coefficients (Lordan et al. 2014), implying high accessibility and efficiency. Further research has highlighted that transport networks are robust to random failures but vulnerable to targeted disruptions at key nodes identified by centrality measures (Abdelaty et al. 2020; Lordan et al. 2014; Sun et al. 2015).

However, only a limited number of studies have extended the application of topological characteristics to examine their relationship with passenger ridership features. For example, Luo et al. (2020) demonstrated that degree, betweenness, and closeness centralities can capture the network context of public transit systems and used them to estimate passenger flows. Dou et al. (2021) incorporated degree and betweenness centralities to improve the classification of metro station functions. These findings highlight the potential of topological features to characterize the broader network context of transport systems and to establish meaningful links with ridership patterns, thereby enhancing the understanding of the relationship between network topology and passenger ridership patterns.

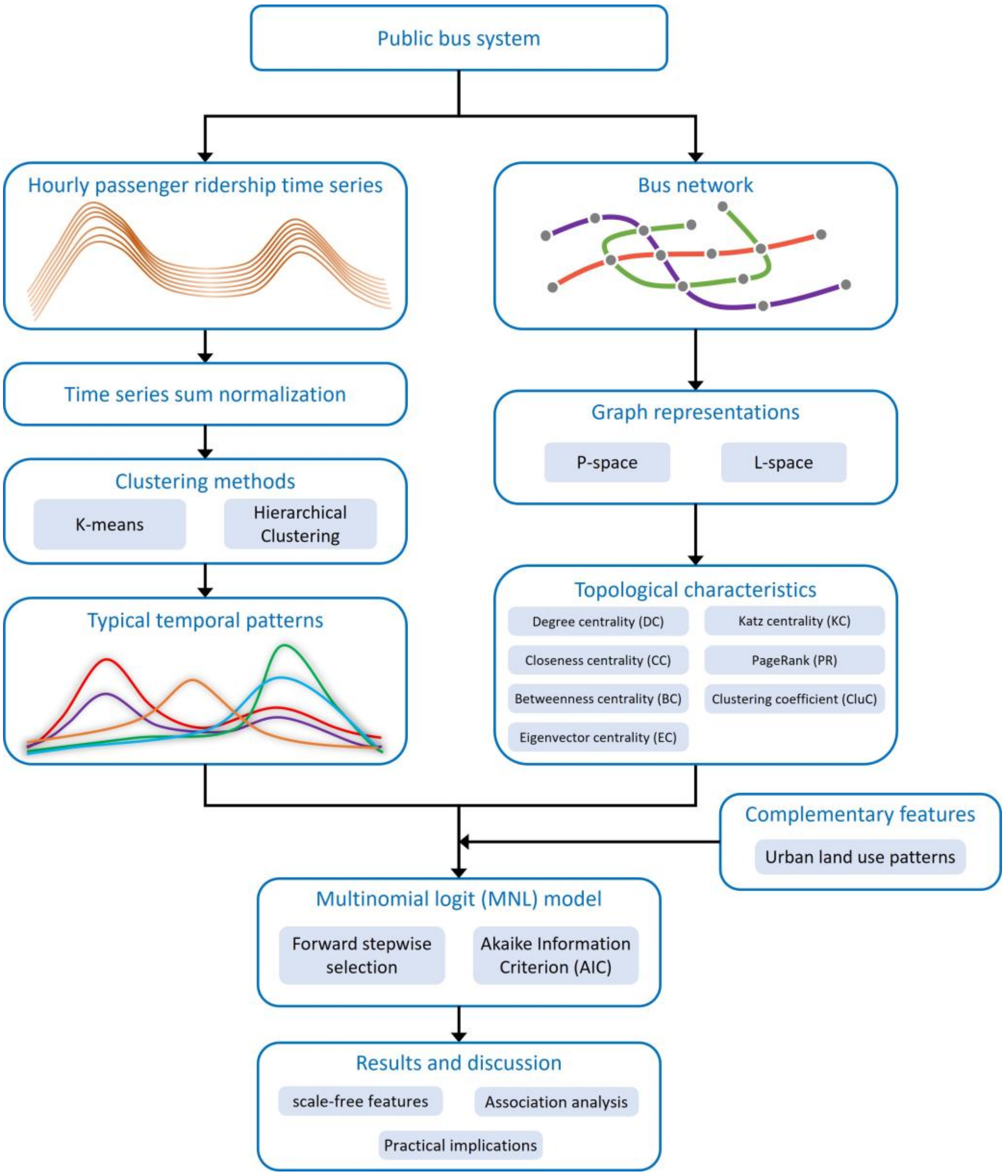


Figure 1. Framework of the study.

3 Methods

Figure 1 presents the framework for examining the relationship between passenger ridership patterns and network topological characteristics. First, hourly passenger boarding volumes at bus stops are collected to form temporal sequences. Sum normalization is employed to expose temporal patterns, and clustering methods are applied to divide the temporal sequences into groups representing typical patterns. Then, the bus network is represented in both L-space and P-space using complex network theory, and corresponding topological characteristics are computed. Finally, an MNL model is employed to analyze the association between ridership patterns and topological features, with independent variables selected through forward stepwise selection based on the Akaike Information Criterion (AIC).

3.1 Clustering analysis

Clustering methods, such as K-means and hierarchical clustering, are employed to partition ridership time series into distinct groups to reveal typical patterns, where members within the same group exhibit similar characteristics, while those in different groups display differences (Choi et al. 2022; Gan et al. 2020; Kim 2018; Kim and Jang 2022; Zhao et al. 2019). Given a dataset $\mathcal{D} = \{\boldsymbol{x}_i | i = 1, \cdots, N\}$ with N samples, a clustering method aims to partition the samples into Q non-overlapping groups, such that $\mathcal{D} = \cup_{q=1}^{Q} \mathcal{D}_q$. The K-means clustering algorithm minimizes the within-cluster sum of squared distances:

$$\min L = \min_{\boldsymbol{c}_q, q=1,\cdots Q} \sum_{q=1}^{Q} \sum_{x \in \mathcal{D}_q} \text{dist}(\boldsymbol{x}, \boldsymbol{c}_q) \tag{1}$$

where $\boldsymbol{c}_q$ is the centroid of cluster $\mathcal{D}_q$, i,e, $\boldsymbol{c}_q = \frac{1}{|\mathcal{D}_q|} \sum_{x \in \mathcal{D}_q} \boldsymbol{x}$, and $|\mathcal{D}_q|$ denoting sample size. $\text{dist}(\boldsymbol{x}, \boldsymbol{c}_q) = \|\boldsymbol{x} - \boldsymbol{c}_q\|^2$ measures the Euclidean distance between $\boldsymbol{x}$ and $\boldsymbol{c}_q$. This optimization problem can be solved using the expectation-maximization (EM) algorithm.

The hierarchical clustering model follows a bottom-up approach, where each sample is initially treated as an individual cluster, and at each iteration, the two most similar clusters are merged. It provides a clear trace of how clusters evolve. Cluster similarity is commonly measured using four linkage criteria: complete linkage, single linkage, average linkage, and Ward's method. Ward's method minimizes the increase in within-cluster variance by merging the two clusters with the smallest between-cluster distance, analogous to the K-means objective function (Ward 1963).

Complete: $$\text{dist}(\mathcal{D}_p, \mathcal{D}_q) = \max_{\boldsymbol{x}_i \in \mathcal{D}_p, \boldsymbol{x}_j \in \mathcal{D}_q} \text{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{2}$$

Single: $$\text{dist}(\mathcal{D}_p, \mathcal{D}_q) = \min_{\boldsymbol{x}_i \in \mathcal{D}_p, \boldsymbol{x}_j \in \mathcal{D}_q} \text{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{3}$$

Average: $$\text{dist}(\mathcal{D}_p, \mathcal{D}_q) = \frac{1}{|\mathcal{D}_p||\mathcal{D}_q|} \sum_{\boldsymbol{x}_i \in \mathcal{D}_p} \sum_{\boldsymbol{x}_j \in \mathcal{D}_q} \text{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{4}$$

All above five clustering models are employed to cluster passenger ridership patterns. The optimal number of clusters, K, is determined using two widely-used metrics: Calinski-Harabasz index (CHI) (Calinski and Harabasz 1974) and Davies-Bouldin index (DBI) (Davies and Bouldin 1979). A higher CHI value and a lower DBI value indicate more compact and well-separated clusters, reflecting superior clustering performance.

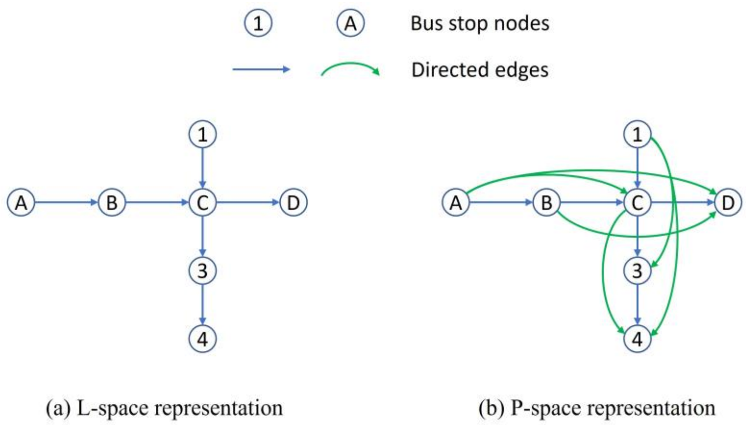


Figure 2 Comparison of two graph models for representing a bus network

3.2 Network topological measurements

Both L-space and P-space models are adopted to represent the bus network, as displayed in Figure 2. The two representations capture different perspectives of the bus network: L-space emphasizes the physical sequencing of stops along routes, while P-space highlights the co-service relationships among bus stops across lines. Specifically, edges reflect the number of hops between stops in L-space but represent the number of line transfers in P-space.

The bus network is represented as an unweighted directed graph $\mathcal{G}(\mathcal{V},\mathcal{E})$, where $\mathcal{V}$ is the set of nodes and $\mathcal{E}$ is the set of edges. Let the number of nodes be $N = |\mathcal{V}|$ and the number of edges is $M = |\mathcal{E}|$. The node and edge sets are denoted as $\mathcal{V} = \{v_i | i = 1,2,3,\cdots,N\}$ and $\mathcal{E} = \{e_{ij} = (v_i, v_j) | v_i, v_j \in \mathcal{V}\}$, where e_{ij} is a directed edge from node v_i to node v_j. The edge set can also be expressed in the form of an adjacency matrix $\boldsymbol{A}$.

$$\boldsymbol{A} = [a_{ij}]_{N\times N}, \qquad \text{where } a_{ij} = \begin{cases} 1, e_{ij} \in \mathcal{E} \\ 0, e_{ij} \notin \mathcal{E} \end{cases} \tag{5}$$

Based on the graph representation of the bus network, seven topological metrics are considered, including Degree Centrality (DC), Closeness Centrality (CC), Betweenness Centrality (BC), Eigenvector Centrality (EC), Katz Centrality (KC), PageRank (PR), and Clustering Coefficient (CluC).

DC is a local metric that quantifies the number of edges directly connected to a node. Nodes with higher DC values are more closely connected and thus considered more important, making it widely used to evaluate the connectivity of transport stations (Lordan et al. 2014; Wu and Kim 2020). In a directed graph, DC consists of in-degree centrality DC_i^{In} and out-degree centrality DC_i^{Out}, representing the numbers of incoming and outgoing edges, respectively, and expressed as:

$$DC_i^{\mathrm{In}} = \Sigma_{j=1}^{N} a_{ji}\,, \qquad DC_i^{\mathrm{Out}} = \Sigma_{i=1}^{N} a_{ij} \tag{6}$$

CC measures the reciprocal of the average shortest path length between a specific node and all others. A higher CC indicates that a node can more easily reach other nodes, thus can reflect the accessibility of transport stations (Wu and Kim 2020). For a node v_i in a directed graph, the shortest path lengths of incoming nodes and outgoing nodes typically are different, and hence, in-CC and out-CC are computed separately, i.e.,

$$CC_i^{\mathrm{In}} = \frac{N-1}{\sum_{v_j\in\mathcal{V}, v_j\neq v_i} d_{ji}^{\min}}, \qquad CC_i^{\mathrm{Out}} = \frac{N-1}{\sum_{v_j\in\mathcal{V}, v_j\neq v_i} d_{ij}^{\min}} \tag{7}$$

where $d_{ji}^{\min}$ denotes the shortest path length from node v_j to node v_i.

BC quantifies the extent to which a node acts as an intermediary along the shortest paths between all node pairs. Nodes with higher BC values serve as bridges that connect different parts of the network. Formally, BC is computed as the sum of the proportions of shortest paths that pass through a given node v_i. Let $d_{st|i}^{\min}$ the number of shortest paths between nodes v_s and v_t that pass through v_i ($d_{st|i}^{\min} = 0$ if none exist). Then,

$$BC_i = \frac{1}{(N-1)(N-2)} \sum_{\substack{i\neq s\neq t,\\ v_i,v_s,v_t\in\mathcal{V}}} \frac{d_{st|i}^{\min}}{d_{st}^{\min}} \tag{8}$$

where N is the number of nodes, and $\frac{1}{(N-1)(N-2)}$ represents the normalization factor for a directed graph.

EC measures a node's importance based on the importance of its neighbors, computed as:

$$EC_i = \frac{1}{\lambda} \Sigma_{j=1}^{N} (a_{ji} \cdot EC_j) \tag{9}$$

where λ is a scaling factor. This recursive formulation can be expressed in matrix form as:

$$\lambda \boldsymbol{x} = \boldsymbol{A}\boldsymbol{x} \tag{10}$$

where $\boldsymbol{A}$ is the adjacency matrix and $\boldsymbol{x} = [EC_1, EC_2, \cdots, EC_N]^{\mathrm{T}}$. By the Perron-Frobenius theorem, λ corresponds to the largest eigenvalue of $\boldsymbol{A}$, and $\boldsymbol{x}$ is the associated eigenvector. For directed graphs, since $\boldsymbol{A}$ is asymmetric, yields the in-EC EC_i^{In}, which considers the importance of incoming nodes. Out-EC EC_i^{Out} is similarly obtained by replacing $\boldsymbol{A}$ is with $\boldsymbol{A}^{\mathrm{T}}$.

KC measures a node's influence by accounting for the total number of walks between pairs of nodes (Katz 1953), computed as:

$$KC_i = \Sigma_{l=1}^{\infty}\Sigma_{j=1}^{N}\alpha^l\left(\boldsymbol{A^l}\right)_{ji} \tag{11}$$

where $\boldsymbol{A^l}$ is the l-th powers of the adjacency matrix $\boldsymbol{A}$, and $\left(\boldsymbol{A^l}\right)_{ji}$ indicates the number of walks of length l from node v_j to node v_i. α is an attenuation factor and must be smaller than the reciprocal of the maximum eigenvalue of $\boldsymbol{A}$. Using the matrix geometric series identity, i.e., $\sum_{l=0}^{\infty}\alpha^l(\boldsymbol{A}^{\mathrm{T}})^{\boldsymbol{l}} = (\boldsymbol{I}-\alpha\boldsymbol{A}^{\mathrm{T}})^{-1}$, KC can be rewritten in matrix form as:

$$KC = ((\boldsymbol{I}-\alpha\boldsymbol{A}^{\mathrm{T}})^{-1}-\boldsymbol{I})\mathbf{1} = (\boldsymbol{I}-\alpha\boldsymbol{A}^{\mathrm{T}})^{-1}\mathbf{1}-\mathbf{1} \tag{12}$$

where $\boldsymbol{I}$ is an identity matrix and $\mathbf{1}$ is an all-one column vector. For directed graphs, Eq. (12) yields in-KC KC_i^{In}, which accounts for incoming edges. Out-KC KC_i^{Out} is obtained by substituting $\boldsymbol{A}^{\mathrm{T}}$ with $\boldsymbol{A}$.

PR (Page et al. 1999) was originally developed by Google to rank web pages, assuming that a node's importance is equally distributed among its outgoing edges, computed as:

$$PR_i = \frac{1-\alpha}{N} + \alpha\sum_{j=1}^{N} a_{ji}\frac{PR_j}{DC_j^{\mathrm{out}}} \tag{13}$$

where α damping factor, typically set between 0.8 to 0.9 in practice. DC_j^{out} denotes the out DC. The equation is solved using the power iteration method (Page et al. 1999). Nodes with higher PR values are regarded as more critical.

CluC measures the extent of clustering among a node and its neighbors (Newman 2010). Its value ranges between 0 and 1. A low CluC indicates that a node functions as a hub, frequently serving as an intermediary when its neighbors connect with each other, whereas a high CluC typically reflects the small-world property. In an unweighted directed graph, CluC is computed as (Fagiolo 2007):

$$CluC_i = \frac{\sum_h\sum_j(a_{ij}+a_{ji})(a_{jh}+a_{hj})(a_{hi}+a_{ih})}{2\left[DC_i^{\mathrm{Total}}\left(DC_i^{\mathrm{Total}}-1\right)-2DC_i^{\leftrightarrow}\right]} \tag{14}$$

where $DC_i^{\mathrm{Total}} = DC_i^{\mathrm{In}} + DC_i^{\mathrm{Out}}$ is the total degree centrality. $DC_i^{\leftrightarrow} = \sum_j a_{ij}a_{ji}$ denotes the number of bilateral edges between node v_i and its neighbors.

3.3 Multinomial logit (MNL) model

MNL model is employed to model the relationship between clustering patterns of ridership time series ($Y_i = j$) and network topological characteristics ($\boldsymbol{x}_i$). The probability of bus stop i being classified into cluster j is expressed as:

$$\Pr(Y_i = j|\boldsymbol{x_i}) = \frac{\exp\left(\beta_{0,j}+\boldsymbol{\beta}_j^{\mathrm{T}}\boldsymbol{x}_i\right)}{\sum_{q=1}^{Q}\exp\left(\beta_{0,q}+\boldsymbol{\beta}_q^{\mathrm{T}}\boldsymbol{x}_i\right)}, \qquad \forall j = 1,2,\cdots,Q \tag{15}$$

where $Y_i = j$ denotes that the ridership pattern of bus stop i belongs to clustering pattern j. $\boldsymbol{x_i}$ is a vector of topological features. Q is the number of clusters. $\beta_{0,j}$ and $\boldsymbol{\beta}_j, j = 1,2,\cdots,Q$ are model coefficients.

The model can be calibrated by maximum likelihood estimation, with one cluster designated as the reference group. If $j = 1$ is chosen as the reference, the relative probability of cluster j compared to the reference is:

$$\begin{aligned}\frac{\Pr(Y_i = j|\boldsymbol{x_i})}{\Pr(Y_i = 1|\boldsymbol{x_i})} &= \exp\left[\left(\beta_{0,j}-\beta_{0,1}\right)+\left(\boldsymbol{\beta}_j-\boldsymbol{\beta}_1\right)^{\mathrm{T}}\boldsymbol{x}_i\right] \\ &= \exp\left(\beta'_{0,j}+\boldsymbol{\beta}'^{\mathrm{T}}_j\boldsymbol{x}_i\right), \qquad \forall j = 2,\cdots,Q\end{aligned} \tag{16}$$

where $\beta'_{0,j}$ and $\boldsymbol{\beta}'_j$ represent coefficients relative to the reference group.

To ensure model performance, the forward stepwise selection strategy based on Akaike Information Criterion (AIC) is employed to determine explanatory variable sets. AIC balances model fit and complexity, with smaller values indicating superior models. The procedure begins with a constant-only model. At each step, the variable yielding the largest reduction in AIC is added. The process continues until no further reduction occurs or all candidate variables are examined.

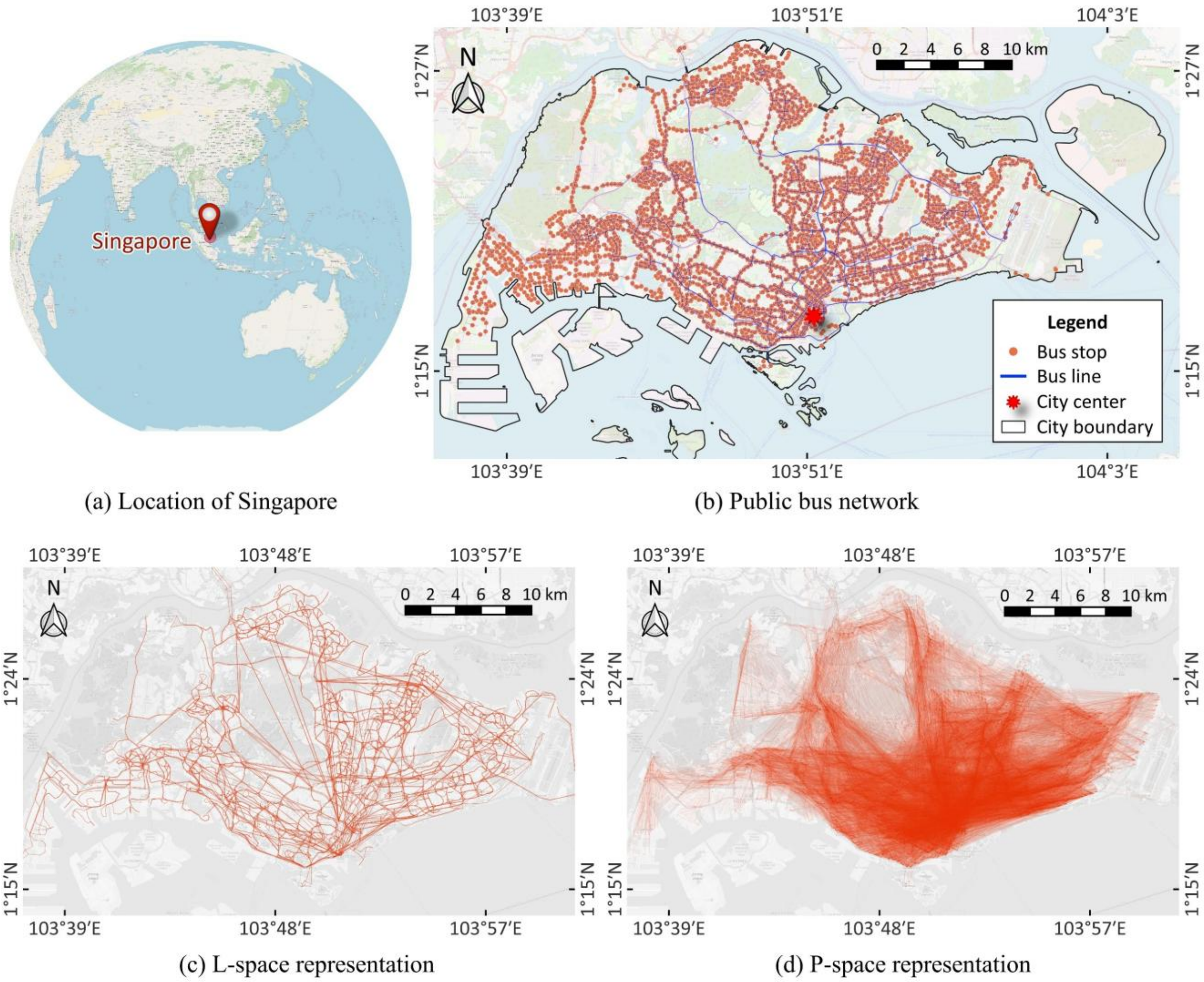


Figure 3. Graph representation of Singapore public bus network

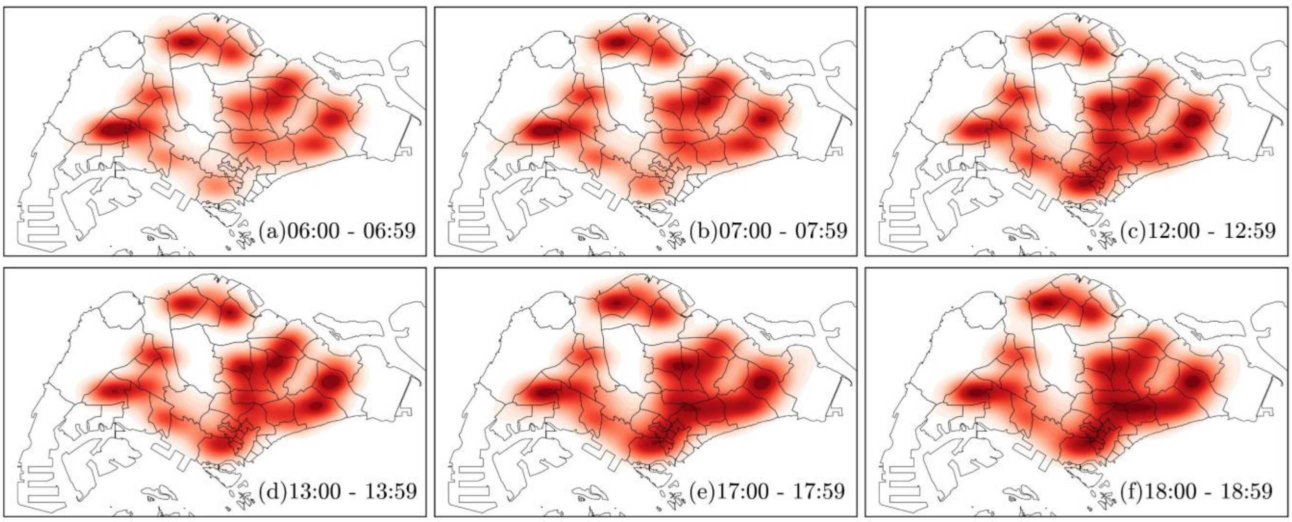


Figure 4. Kernel density plots indicating spatial distribution of hourly passenger volume.

4 Study Case

4.1 Study area

Singapore is examined as a case study. It is a city-state in Southeast Asia, located off the southern tip of the Malay Peninsula, as shown in Figure 3(a). With a population of approximately five million residing within an area of about 710 square kilometers, Singapore is the third most densely populated country in the world. Public transport systems (i.e., buses and metros) are strongly promoted. In 2020, public transit accounted for 57.7% of daily trips, more than double the share of private transport modes (e.g., cars and taxis) (Singapore Department of Statistics 2021). As of October 2022, Singapore's bus system is jointly operated by four companies: SBS Transit, SMRT, Go-Ahead Singapore, and Tower Transit Singapore. Figure 3(b) displays the geospatial layout of the bus network. The bus system comprises 698 directed bus lines and 5,075 bus stops, covering 4,866 kilometers of urban roadways and serving approximately 5.6 million passenger trips on an average workday.

4.2 Data collection and processing

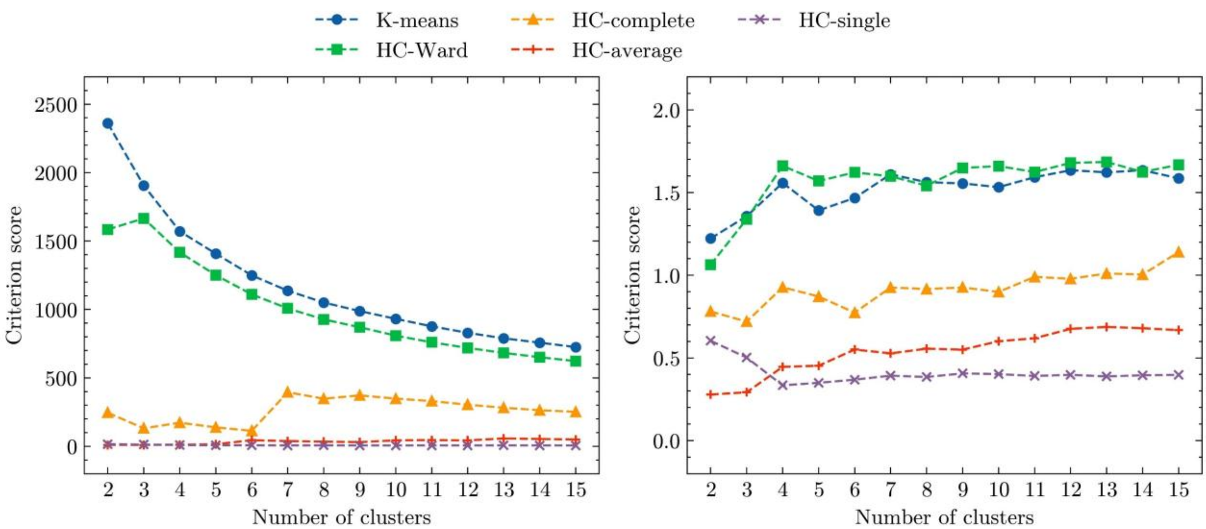
Data on bus stops, bus service routes, and passenger boarding volumes were obtained from Singapore Land Transport Authority (LTA) DataMall platform for October 2022. The bus stop and service route data were used to construct network graphs. Figure 3(c) and (d) present the L-space and P-space representations of the bus network, respectively. The P-space graph appears denser than the L-space because it introduces additional edges between all pairs of nodes served by the same bus route. Then, average hourly boarding volumes on workdays were used to extract ridership time-series patterns for each bus stop. To avoid potential outliers, records outside regular operating hours (i.e., 6:00 - 24:00, 18 hours) and bus stops with low demand (daily volume < 100) were excluded. An 18-step passenger volume time series was obtained for each remaining bus stop. Figure 4 presents the spatial distribution of hourly boarding volume through kernel density plots, highlighting the spatiotemporal dynamics of daily bus trips. Finally, sum normalization, dividing each hourly volume by the corresponding daily total, was applied to exposure ridership patterns (Pei et al. 2014).

5 Results and Discussions

5.1 Clustering analysis of passenger ridership time series patterns

Figure 5 compares the clustering criterion score of the five different clustering methods across varying cluster numbers. The results show that the K-means approach performs best in terms of CHI and performs comparably to hierarchical clustering with Ward's linkage in terms of DBI. Although both indices suggest two clusters as optimal, this solution is overly generalized and does not adequately capture ridership variability. To further assess the appropriate number of clusters, the sensitivity of cluster centroids across different values is analyzed, as displayed in Figure 6. When the number of clusters is set to two, only morning and afternoon peak patterns emerge. Notably, when the number increases to five, a distinct group with a noon peak emerges. Beyond five clusters, no additional meaningful patterns appear. Therefore, the K-means model with five clusters is adopted to identify typical ridership patterns.

Table 1 summarizes the shares of passenger volume during the morning, noon, and afternoon peak periods. These peak patterns reflect aggregated passenger trip characteristics and the functional roles of bus stops. Figure 7 illustrates the spatial distribution of the five clusters using kernel density plots, along with their typical temporal patterns. Figure 8 summarizes the land use composition around bus stops, based on 200-meter and 400-meter buffers, providing key supporting evidence to analyze bus stop functions (Gan et al. 2020; Liu et al. 2019).



(a) Calinski-Harabasz index (CHI) (b) Davies-Bouldin index (DBI)

Figure 5. Clustering performance of different clustering methods and cluster numbers

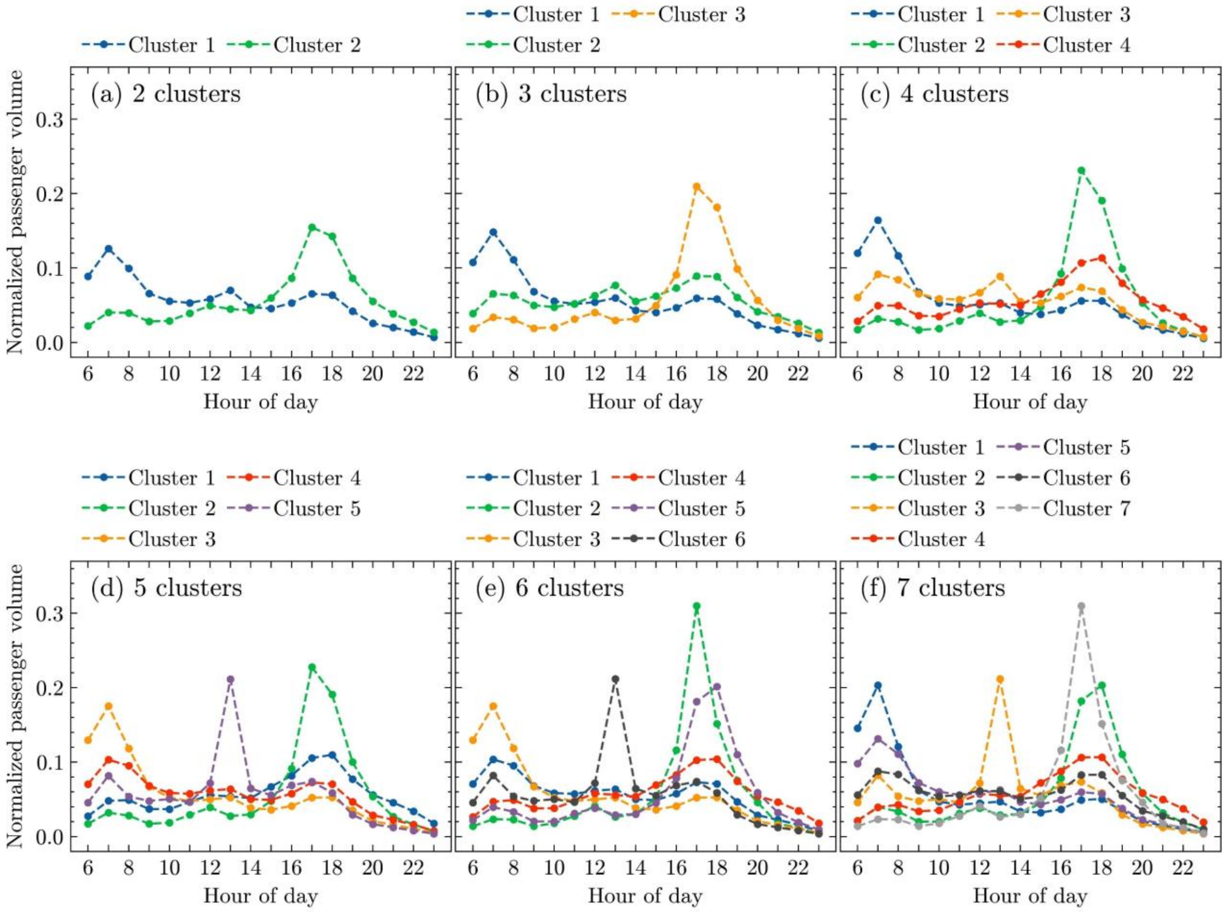


Figure 6 Sensitivity analysis of the cluster number of K-means method.

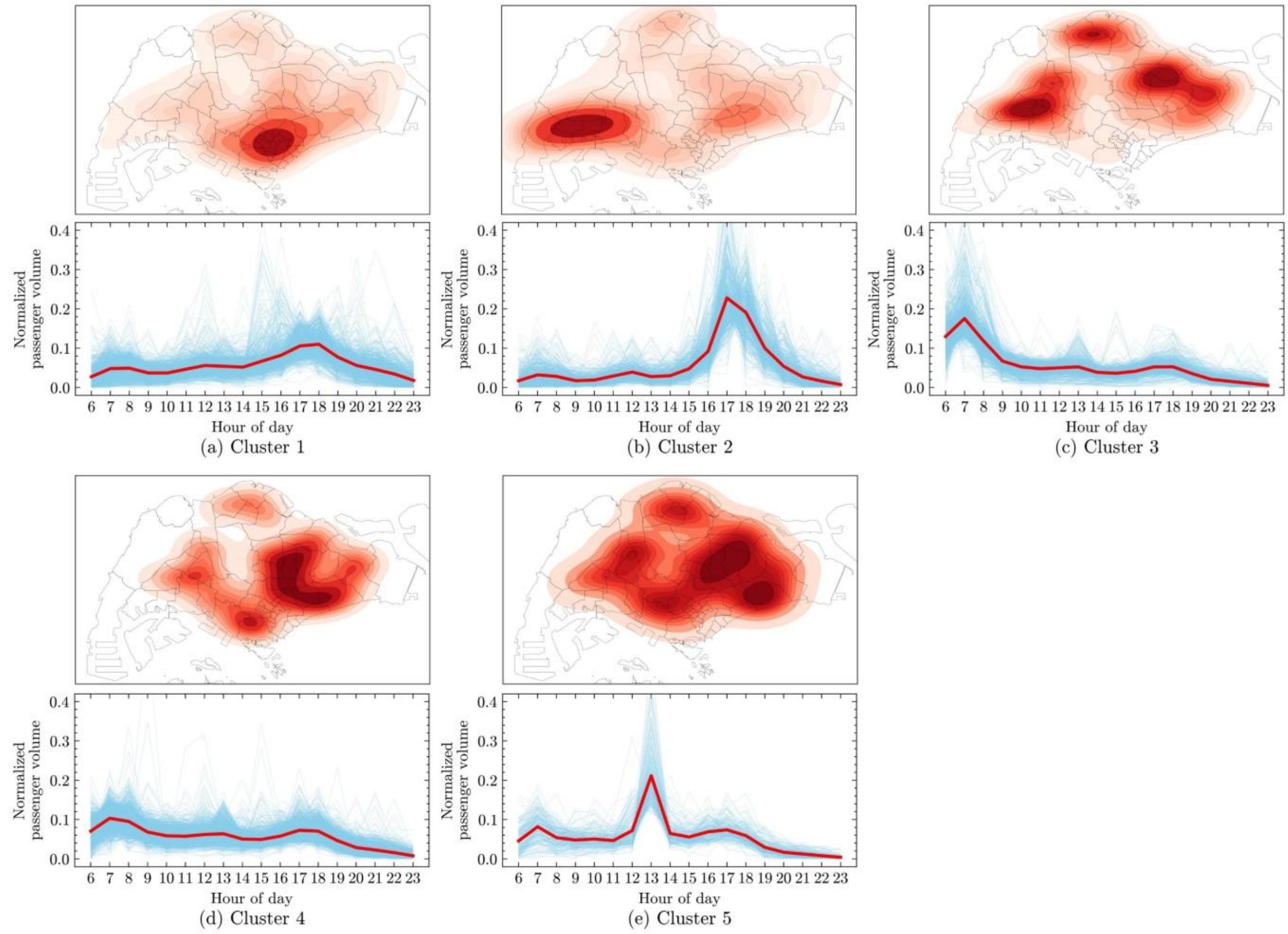


Figure 7 Kernel density plots indicating spatial distributions of the five clusters and their typical temporal patterns.

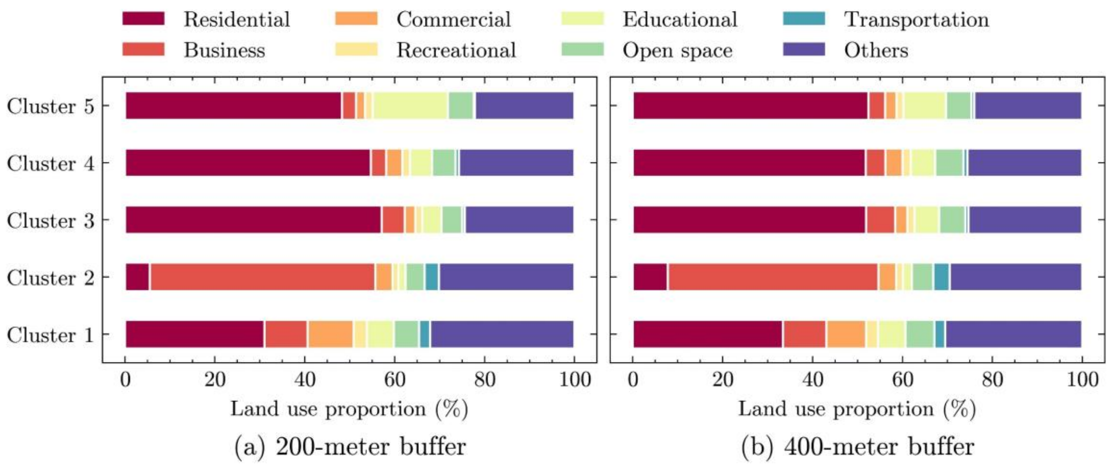


Figure 8. Land use composition surrounding bus stops

Table 1. Characteristics of peak hour ridership volume.

	Cluster 1	**Cluster 2**	**Cluster 3**	**Cluster 4**	**Cluster 5**
Number of bus stops	1054	593	1061	1620	227
Median of daily volume	6015	2504	7738	6852	4773
Percentage of morning peak (6:00-8:00) volume	7.53%	4.93%	30.49%	17.36%	12.72%
Percentage of noon peak (12:00-14:00) volume	10.97%	6.66%	10.23%	12.56%	28.33%
Percentage of afternoon peak (17:00-19:00) volume	21.53%	41.87%	10.48%	14.32%	13.26%
Morning-to-afternoon peak volume ratio	34.99%	11.77%	290.93%	121.27%	95.89%

The explanation of each cluster is summarized as follows:

Cluster 1: Mixed-use city business district (CBD). Temporally, Clusters 1 and 2 both exhibit higher boarding passenger volumes during the afternoon peak than during the morning peak, reflecting work-related trips departing from workplaces in the evening. For Cluster 1, the afternoon peak accounts for 21.53% of the daily boarding volume, approximately three times the morning peak. Spatially, Cluster 1 is primarily concentrated in the city center, corresponding to CBD areas. Additionally, Cluster 1 has the largest proportion of commercial land (7.10% within 200-meter buffers) among all clusters and the second-largest proportion of business land (11.26% within 200-meter buffers), suggesting a mixed commercial-business land use environment. This mixed-use composition likely results in less pronounced evening peaks than those observed in much denser employment regions (i.e., Cluster 2).

Cluster 2: Western job hubs. Temporally, Cluster 2 exhibits highest afternoon peaks among all clusters, accounting for 41.87% of the daily boarding volume and nearly nine times the morning peak. Spatially, Cluster 2 is concentrated in the western regions of Singapore, which are characterized by employment-intensive areas with lots of industrial and manufacturing factories. Additionally, the surrounding environment is dominated by business lands (51.20% within 200-meter buffers), indicating the highest employment density.

Cluster 3: Peripheral residential regions. Temporally, Clusters 3 and 4 experience higher boarding volumes during the morning peak than during the afternoon peak. This indicates home-based commuting as residents typically depart from their homes in the morning. Cluster 3 shows the strongest morning peak among all clusters, with 30.49% of the daily boarding volume, approximately three times its afternoon peak. Spatially, Cluster 3 is primarily located in the urban periphery. The surrounding environment is dominated by residential land (57.16% within a 200 m buffer), emphasizing the predominance of home-origin trips.

Cluster 4: Near-central residential regions. Temporally, Cluster 4 displays a more moderate morning peak than Cluster 3, with morning volume accounting for 17.36% of daily trips, only slightly higher than the afternoon peak. Spatially, Cluster 4 is located closer to the inner city compared to Cluster 3. The surrounding environment is dominated by residential land (55.25% within a 200 m buffer), the second highest among all clusters, which suggests a prevalence of home-based bus trips.

Cluster 5: School-oriented zones. Cluster 5 is the smallest group with 227 bus stops. Temporally, this cluster exhibits a pronounced noon peak, nearly twice the magnitude of either the morning or afternoon peak. Spatially, Cluster 5 is more dispersed than the other clusters. This cluster has the largest proportion of educational land use (16.71% within 200-meter buffers), potentially associated with student dismissal trips at the end of the school day.

Table 2. Topological characteristics of the bus network. "Used" column suggests the included variables in the MNL model.

	L-space			P-space		
	Average	Std.	Used	Average	Std.	Used
In-degree centrality DC^{In}	1.46	0.79	√	80.57	85.06	√
Out-degree centrality DC^{Out}	1.46	0.79	√	80.57	84.99	√
Closeness centrality CC	4.04×10^{-2}	7.47×10^{-3}	√	3.50×10^{-1}	4.03×10^{-2}	
Betweenness centrality BC	4.92×10^{-3}	9.19×10^{-3}	√	3.74×10^{-4}	2.84×10^{-3}	√
In-eigenvector centrality EC^{In}	3.09×10^{-3}	1.37×10^{-2}	√	8.95×10^{-3}	1.08×10^{-2}	√
Out-eigenvector centrality EC^{Out}	2.96×10^{-3}	1.37×10^{-2}	√	8.95×10^{-3}	1.08×10^{-2}	√
In-Katz centrality KC^{In}	1.74×10^{-1}	9.82×10^{-2}		9.26×10^{-2}	9.91×10^{-2}	
Out-Katz centrality KC^{Out}	1.74×10^{-1}	9.82×10^{-2}		9.26×10^{-2}	9.90×10^{-2}	
PageRank PR	1.97×10^{-4}	9.50×10^{-5}	√	1.97×10^{-4}	2.66×10^{-4}	√
Clustering coefficient $CluC$	7.63×10^{-3}	3.96×10^{-2}	√	3.47×10^{-1}	1.66×10^{-1}	√
Number of nodes N	5,075			5,075		
Number of edges M	7,383			408,894		
Network density $M/(N^2-N)$	2.87×10^{-4}			1.59×10^{-2}		
Network diameter $\max_{v_i,v_j\in\lvert\mathcal{V}\rvert} d_{ij}^{\min}$	118			6		
Average shortest path length $\left(\sum_{i\neq j} d_{ij}^{\min}\right)/((N-1)(N-2))$	25.96			2.90		

5.2 Analysis of bus network topological characteristics

Table 2 compares the topological characteristics of the bus network in both L-space and P-space. The number of edges in P-space is nearly 55 times greater than that in L-space. The average DC exhibits the same values, since the sum of all nodes' in-degree or out-degree centralities is equivalent to the total number of edges. The interpretation of shortest path length differs between the two spaces: in L-space, it represents the number of intermediate bus stops encountered during a journey, whereas in P-space, it reflects the number of transfers required. Accordingly, the network diameter suggests that the longest possible journey in the system involves traversing 118 bus stops in L-space or making 6 transfers in P-space.

5.2.1 Degree distributions

Public transit networks have been frequently reported to exhibit power-law degree distributions, typically expressed as $P(k)\sim k^{-\gamma}$ (Abdelaty et al. 2020; Huang et al. 2015). When plotted on a logarithmic scale, such distributions appear approximately linear (see Figure 9). The power-law property indicates that the network is dominated by a minority of high-degree nodes (i.e., hubs) while the majority of nodes are connected to these hubs. In real-world bus networks, these hubs typically correspond to bus interchanges where multiple bus lines intersect. Consequently, transport hubs are often identified as critical or vulnerable nodes, warranting particular attention in network analysis and transport planning (Huang et al. 2015).

A network is typically regarded as scale-free if its degree distribution follows a power law with an exponent γ between two and three. As shown in Figure 9 (a) and (b), both the in-degree and out-degree distributions in L-space conform well to a power-law form, with relatively strong goodness-of-fit R^2. However, the corresponding power exponents exceed three, thereby rejecting the scale-free assumption, a common finding in public transit networks. For instance, Sienkiewicz and Hołyst (2005) examined transit systems in 22 Polish cities and found that 15 exhibited exponents greater than three. Huang et al. (2015) reported that the exponent for the Beijing public transit network was close to four. Nevertheless, even with exponents above three, transport networks may still display small-world properties, including a short average path length and a high clustering coefficient (Barabási and Pósfai 2016). In P-space in Figure 9 (c) and (d), the degree distribution is instead fitted with a truncated power law. Because nodes with low degrees deviate from the straight-line relationship, those with degree values below ten were excluded from the analysis. The resulting fits show that both in-degree and out-degree distributions in P-space are weaker than those observed in L-space, consistent with Lin and Ban's study (2013).

5.2.2 Correlation analysis of topological characteristics

Figure 10 illustrates the distribution of topological features. Overall, the variables in P-space and L-space exhibit similar distributional shapes. Most distributions are highly skewed with long tails, resembling the behavior observed in the degree distribution. The skewness is more pronounced in L-space, indicating that the majority of nodes fall within a narrow range of values. This is reasonable, as most bus stops function as intermediate points along a bus line, each typically connected to one predecessor and one successor. In contrast, in P-space these intermediate nodes are connected by a greater number of edges, resulting in more diverse topological characteristics and consequently a broader distribution range.

To reduce skewness and facilitate MNL modeling, a logarithmic transformation was applied to all topological variables. Subsequently, Pearson correlation coefficients were calculated to examine potential multicollinearity among the transformed indicators, presented in Figure 11. In L-space, strong linear relationships are observed between two pairs of variables: DC^{In} and KC^{In}, DC^{Out} and KC^{Out}. Their correlations are even more severe in P-space, with coefficients approaching unity. Moreover, in P-space, a strong linear relationship is also identified between EC^{In} and CC. To address the issue of multicollinearity, KC^{In} and KC^{Out} were removed in both P-space and L-space, while CC is further dropped in P-space. The final variable set retained for modeling is indicated by the "Used" columns in Table 2.

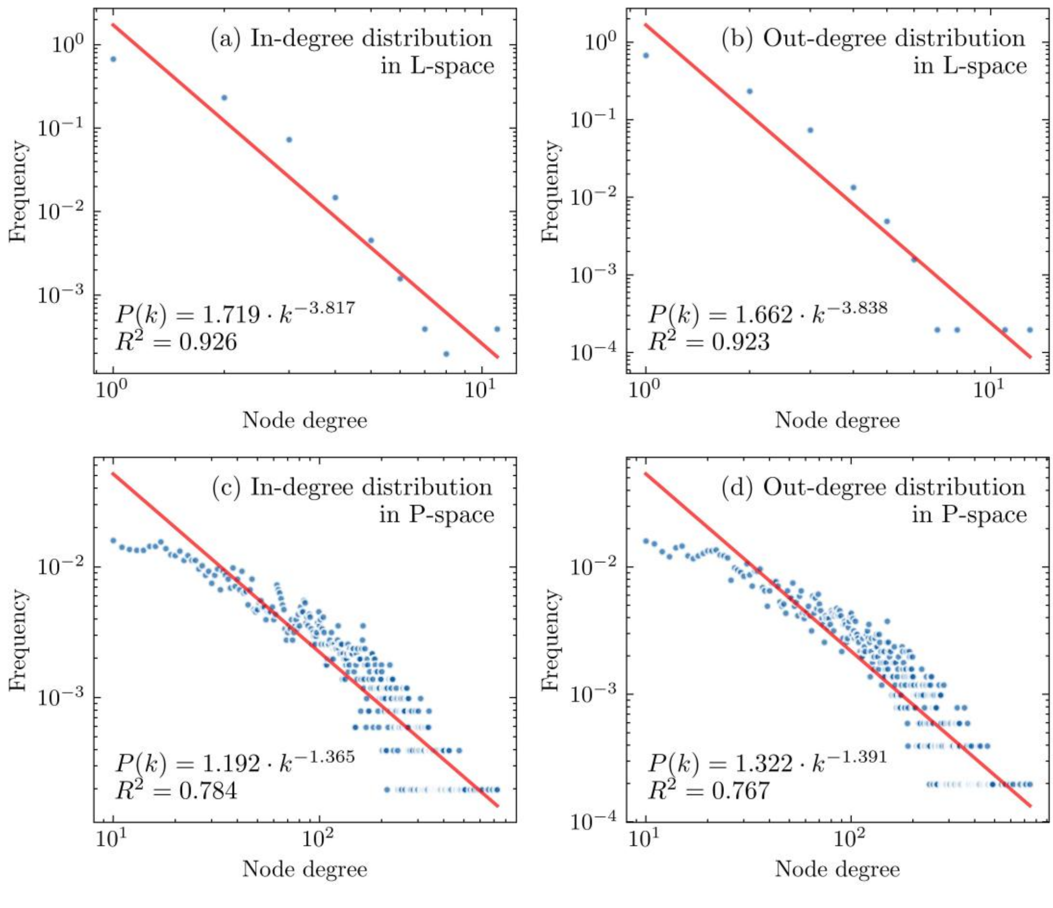


Figure 9. Degree distribution

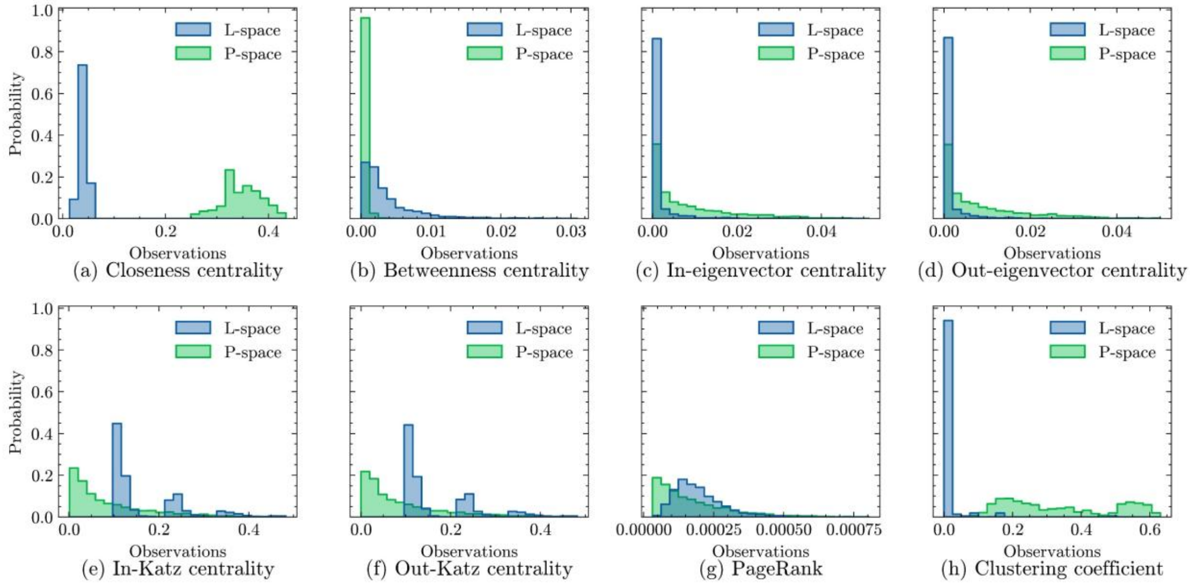


Figure 10. Distribution of topological measurements

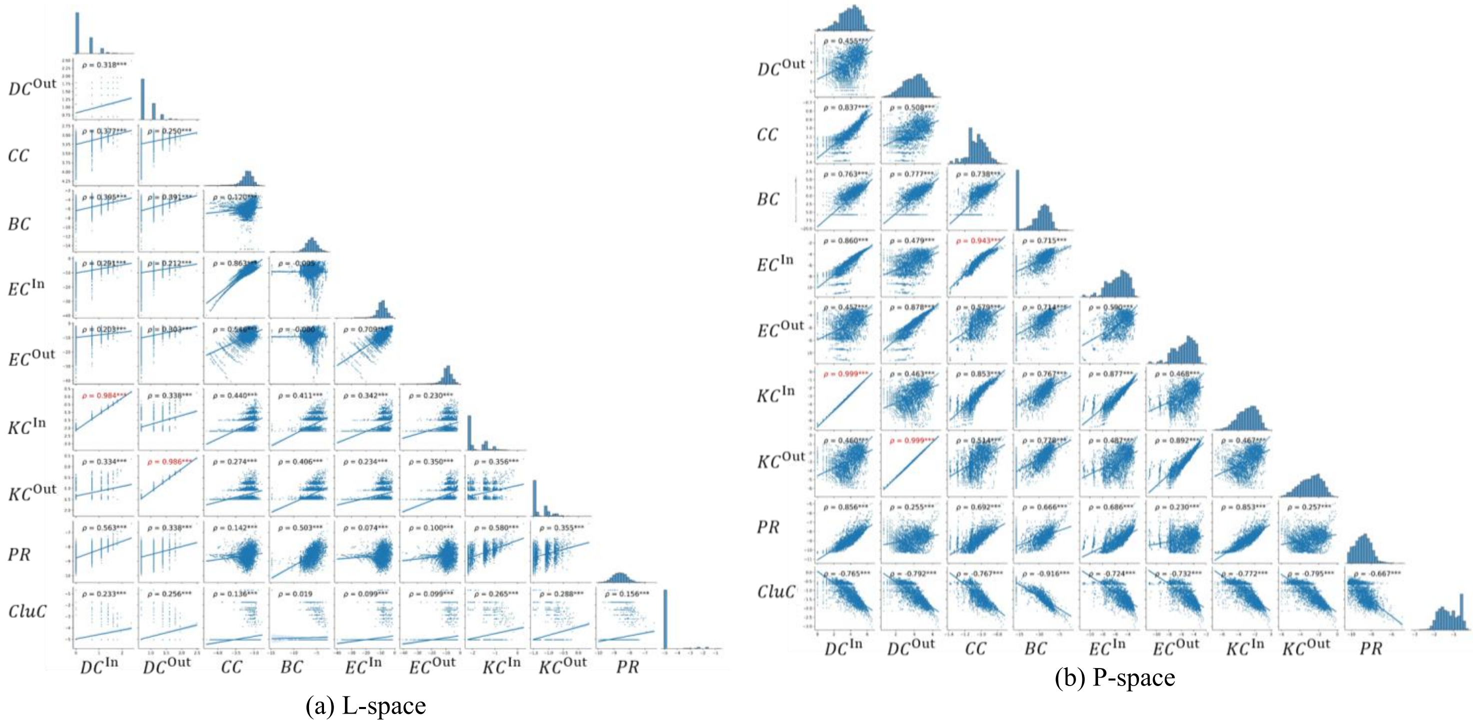


Figure 11. Pearson correlation of network topological features.

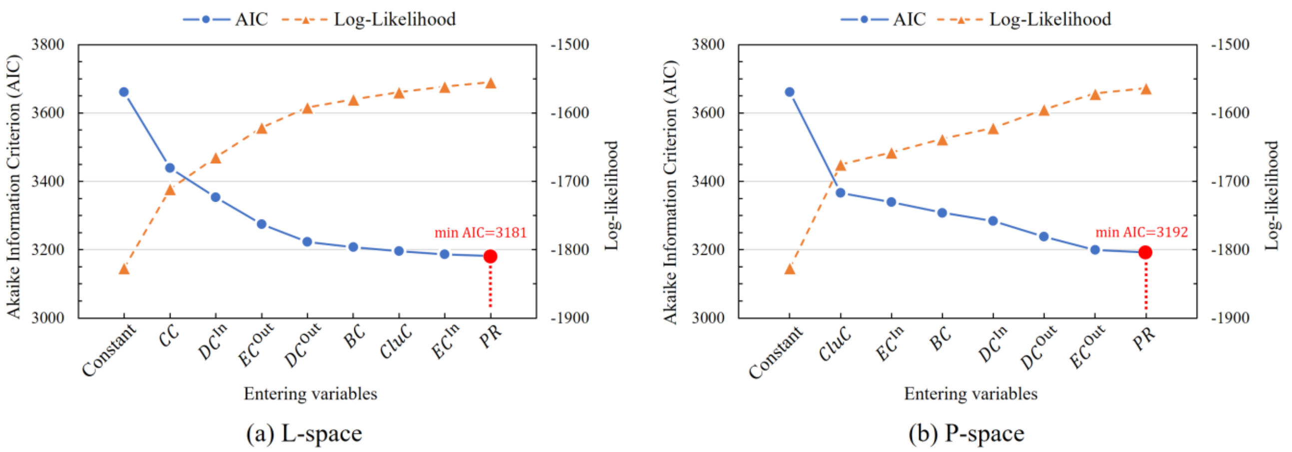


Figure 12. Forward stepwise selection based on AIC

5.3 Relationship between ridership clustering patterns and bus stop topological characteristics

The relationship between ridership clustering patterns and their topological characteristics was examined using an MNL model. To mitigate potential influence from class imbalance, a down-sampling process was applied so that each class had the same sample size as the minority class. Specifically, samples farthest from the class centroid were removed. Following this, a forward stepwise selection method based on the Akaike Information Criterion (AIC) was employed to determine the optimal set of explanatory variables, as displayed in Figure 12. The AIC drops sharply with the inclusion of the first variable, then declines more gradually as additional variables are added, reaching its minimum when all variables are included.

Table 3 and Table 4 provide the estimation results for models incorporating topological features in L-space and P-space, respectively, specifically, including five model specifications: L-space only (Model 1), P-space only (Model 2), land use only (Model 3), L-space and land use (Model 4), as well as P-space and land use (Model 5). For the topology-only specifications (Models 1 and 2), both models exhibit statistically significant improvements over the constant-only model, suggesting a clear association between temporal volume patterns and topological characteristics. Most coefficients for Cluster 2 and Cluster 3 (and for Cluster 5 in L-space) are statistically significant, suggesting their relative temporal profiles can be meaningfully distinguished through topological features. Whereas, coefficients for Cluster 4 are largely insignificant relative to Cluster 1, likely because these two clusters display similar temporal profiles, with relatively moderate peak patterns (see Figure 7(a) and (d)), and similar surrounding network configurations. Notably, the inclusion of land use features appears to better capture differences between Cluster 1 and Cluster 4 (Models 3, 4, 5).

In L-space (Model 1) in Table 3, DC^{In} is negative for Cluster 2, 4, and 5, whereas DC^{Out} is negative for Cluster 2 and 5 but positive for Cluster 3. Higher in-degree typically characterizes stops located at convergence points where multiple upstream segments merge. The negative coefficients therefore suggest that these clusters are less likely to be located at such convergence points. This pattern is plausible for employment hubs (Cluster 2) and school-oriented zones (Cluster 5), which generally function as destination terminals rather than interchanges. DC^{Out} is positive only for Cluster 3, supporting its interpretation as a residential-origin cluster. These stops tend to exhibit relatively stronger outward connectivity to multiple downstream destinations. CC is negative and strongly significant for Cluster 2, 3, and 5, indicating that Cluster 1 constitutes the most topologically central category, exhibiting shorter average distances to the rest of the network. BC is strongly positive for Cluster 2, reflecting its role as bridging nodes along many shortest paths. EC^{In} is positive for Cluster 2, 3, and 5 and EC^{Out} is negative for the same clusters, suggesting that these clusters tend to receive more flows from more influential upstream nodes but are less connected to equally influential downstream nodes. Such a configuration is consistent with destination-oriented roles, including job centers and schools, which attract inflows from major corridors but redistribute less further downstream. PR is significantly negative only for Cluster 2, suggesting employment hubs may distribute flows across many outward directions. $CluC$ is negative for Clusters 3, 4, and 5, exhibiting their lower local transitivity relative to Cluster 1.

In P-space (Model 2) in Table 4, the estimation results exhibit patterns distinct from those obtained in L-space, reflecting the dense co-service structure of route cliques. DC^{In} is positive for Cluster 2 and negative for Cluster 3, whereas DC^{Out} is negative for Cluster 3 and Cluster 4, suggesting job places (Clusters 1 and 2) are better served by long and heavily served routes, compared to residential zones (Clusters 3 and 4). This reflects the spatial distribution of bus lines, in which employment centers are concentrated in urban cores and served by multiple overlapping trunk bus lines, whereas residential areas are more dispersed and primarily connected through branch services. BC is negative for Cluster 2 but positive for Cluster 3, implying that employment regions are not singled out as bridges in P-space (i.e., redundant shortest paths exist within cliques), whereas stops in residential regions act as connectors between lines or between cliques. EC^{In} is negative for Cluster 2 and 3, reduced upstream influence once co-service disperses importance across entire lines. By contrast, EC^{Out} is positive for Cluster 3 and 4, suggesting stronger transmission of importance to influential downstream neighbors. $CluC$ is strongly positive for Cluster 2, 3, and 5, indicating high local transitivity from route cliques.

In summary, the widely used land use features offer more intuitive insights into urban functions when explaining ridership patterns. Importantly, model performance improves when topological features are incorporated. Relative to the land use-only specification, the inclusion of L-space topological features increases the adjusted pseudo-R2 by approximately 13%, to 0.414, while the inclusion of P-space topological features yields an improvement of approximately 15%, to 0.420. These results suggest the topological features provide a complementary perspective to land use features by capturing the broader network context, such as connectivity and transfer potential, thereby enhancing the understanding of temporal passenger volume profiles within public bus systems (Dou et al. 2021).

Table 3. MNL estimation results of L-space only, L-space and land use, and land use only models.

	Model 1: L-space only				**Model 4: L-space and land use**				**Model 3: land use only**			
Variables	**Cluster 2**	**Cluster 3**	**Cluster 4**	**Cluster 5**	**Cluster 2**	**Cluster 3**	**Cluster 4**	**Cluster 5**	**Cluster 2**	**Cluster 3**	**Cluster 4**	**Cluster 5**
Constant	-28.167***	-20.784***	-2.620	-14.146***	4.956	-27.486***	-8.265*	-20.720***	0.157	-5.311***	-2.481***	-3.653***
	(5.319)	(4.933)	(4.637)	(4.845)	(7.779)	(5.711)	(5.003)	(5.860)	(0.567)	(0.908)	(0.676)	(0.785)
DC^{In}	-1.147***	-0.406	-0.638**	-1.720***	-1.457***	-0.090	-0.402	-1.539***				
	(0.333)	(0.287)	(0.264)	(0.314)	(0.498)	(0.332)	(0.286)	(0.368)				
DC^{Out}	-2.497***	-0.867**	0.130	-1.512***	-3.148***	-0.067	0.723*	-0.796				
	(0.494)	(0.391)	(0.368)	(0.449)	(0.725)	(0.454)	(0.396)	(0.518)				
CC	-8.377***	-6.432***	-1.453	-3.805***	-1.964	-6.921***	-2.360*	-4.562***				
	(1.390)	(1.306)	(1.271)	(1.305)	(2.001)	(1.465)	(1.338)	(1.506)				
BC	0.501***	0.197*	0.041	-0.084								
	(0.130)	(0.118)	(0.111)	(0.108)								
EC^{In}	0.154**	0.254***	0.084	0.230***	-0.177	0.318***	0.154**	0.263***				
	(0.074)	(0.070)	(0.070)	(0.071)	(0.111)	(0.083)	(0.076)	(0.083)				
EC^{Out}	-0.116**	0.157***	-0.033	-0.109*	0.123	0.097	-0.089	-0.172***				
	(0.057)	(0.059)	(0.057)	(0.057)	(0.080)	(0.071)	(0.064)	(0.066)				
PR	-0.777**	-0.446	0.323	-0.278	0.935*	-0.187	0.401	-0.434				
	(0.335)	(0.306)	(0.289)	(0.308)	(0.491)	(0.340)	(0.295)	(0.344)				
$CluC$	-0.160	-0.471***	-0.356***	-0.476***	0.033	-0.391**	-0.308**	-0.432**				
	(0.170)	(0.143)	(0.121)	(0.175)	(0.237)	(0.159)	(0.131)	(0.209)				
Residential					-6.347***	9.744***	5.316***	3.205***	-5.693***	9.252***	4.869***	4.668***
					(1.128)	(1.236)	(0.926)	(0.986)	(1.011)	(1.211)	(0.929)	(1.056)
Business					7.330***	3.776	2.629*	4.391***	7.032***	2.680	2.198	6.351***
					(1.345)	(2.375)	(1.589)	(1.479)	(1.193)	(2.337)	(1.565)	(1.457)
Commercial					-2.067	-13.809***	-2.656	-7.287***	-3.506**	-15.059***	-3.198*	-5.853**
					(1.729)	(3.540)	(1.667)	(2.668)	(1.509)	(3.486)	(1.658)	(2.454)
Recreational									-3.189	-3.849	-3.136	2.475
									(2.189)	(3.563)	(2.534)	(1.996)
Educational					4.134*	4.096**	3.342*	14.076***	3.564	4.907**	2.941*	15.746***
					(2.132)	(1.997)	(1.723)	(1.675)	(2.235)	(1.955)	(1.763)	(1.737)
Open space					1.477	6.880***	3.575**	4.675***	2.249	6.435***	2.953*	6.014***
					(1.752)	(1.914)	(1.634)	(1.676)	(1.487)	(1.821)	(1.638)	(1.630)
Transportation					-0.536	-5.011	-15.981**	-42.123***	0.337	-7.198	-12.487**	-40.095***
					(1.551)	(6.936)	(6.792)	(12.746)	(1.335)	(7.661)	(6.207)	(12.681)
Number of regressors	9				14				8			
Degree of freedom	32				52				28			
AIC	3181				2140				2321			
BIC	3362				2422				2482			
Log-likelihood: $LL(\beta)$	-1555				-1014				-1128			
Pseudo R^2:	0.149				0.445				0.382			
Adjusted pseudo R^2	0.129				0.414				0.365			
χ^2 statistic	544				1626				1397			
p-value	<0.001				<0.001				<0.001			

Notes: (1) Cluster 1 as the reference level. (2) ***, **, and * indicate significance levels of $p < 0.01$, $p < 0.05$, and $p < 0.1$, respectively. (3) Estimation standard errors of coefficients in parentheses. (4) Log-likelihood of the null model (i.e., constant-only model): $LL(c) = -1827$.

Table 4. MNL estimation results of P-space only, and P-space and land use models.

	Model 2: P-space only				Model 5: P-space and land use			
Variables	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 2	Cluster 3	Cluster 4	Cluster 5
constant	-7.804	27.940***	3.528	3.394	-2.991	20.231***	-3.559	-6.456
	(5.269)	(5.186)	(5.195)	(5.187)	(8.578)	(6.507)	(6.104)	(6.592)
DC^{In}	1.095***	-0.711**	0.301	-0.055	0.664	-0.679*	0.361	0.186
	(0.378)	(0.344)	(0.370)	(0.352)	(0.522)	(0.403)	(0.401)	(0.410)
DC^{Out}	0.394	-1.952***	-0.866**	-0.445	-0.394	-1.657***	-0.451	-0.045
	(0.384)	(0.390)	(0.399)	(0.375)	(0.570)	(0.470)	(0.453)	(0.453)
BC	-0.305***	0.371***	-0.153	-0.045	-0.262	0.379***	-0.172	-0.030
	(0.118)	(0.116)	(0.120)	(0.113)	(0.173)	(0.137)	(0.132)	(0.133)
EC^{In}	-0.557**	-0.611***	-0.129	-0.004	-0.363	-0.737***	-0.208	-0.109
	(0.231)	(0.222)	(0.249)	(0.232)	(0.322)	(0.268)	(0.276)	(0.278)
EC^{Out}	0.219	1.376***	0.921***	0.307	0.556	1.263***	0.694**	-0.060
	(0.242)	(0.256)	(0.270)	(0.238)	(0.370)	(0.315)	(0.305)	(0.297)
PR	-0.088	0.725**	-0.238	-0.321	-0.319	0.693*	-0.209	-0.516
	(0.339)	(0.299)	(0.289)	(0.330)	(0.564)	(0.359)	(0.324)	(0.395)
$CluC$	2.845***	2.108***	0.360	2.085***	1.204	1.518**	-0.113	1.464*
	(0.696)	(0.642)	(0.615)	(0.657)	(1.057)	(0.768)	(0.695)	(0.788)
Residential					-6.408***	7.582***	5.263***	2.752**
					(1.127)	(1.320)	(1.028)	(1.074)
Business					6.309***	0.373	2.528	4.556***
					(1.323)	(2.699)	(1.662)	(1.543)
Commercial					-3.419**	-17.754***	-2.914*	-6.901***
					(1.601)	(3.924)	(1.738)	(2.612)
Recreational					-3.877*	-7.508*	-2.712	1.001
					(2.067)	(3.911)	(2.630)	(1.871)
Educational					1.807	2.352	3.262*	12.909***
					(2.244)	(2.121)	(1.761)	(1.714)
Open space					1.486	4.640**	3.932**	4.307**
					(1.744)	(2.080)	(1.736)	(1.776)
Transportation					-0.214	-11.722	-9.388	-37.094***
					(1.553)	(8.728)	(5.917)	(12.779)
Number of regressors	8				15			
Degree of freedom	28				56			
AIC	3192				2117			
BIC	3353				2419			
Log-likelihood: $LL(\beta)$	-1564				-999			
Pseudo R^2:	0.144				0.453			
Adjusted pseudo R^2	0.126				0.420			
χ^2 statistic	526				1656			
p-value	<0.001				<0.001			

5.4 Policy implications

The findings indicate that Cluster 2 (western job hubs) exhibits the sharpest evening peak in boarding volumes, highlighting a heightened risk of localized crowding and delay propagation. This cluster is primarily associated with labor-intensive employment sectors, like manufacturing or industrial factories, and serves a workforce with a high reliance on public transit. Compared with CBD areas (Cluster 1), these bus stops are characterized by significantly lower levels of network connectivity and structural cohesion, which may result in relatively underserved bus services and a greater vulnerability to crowding and congestion during peak periods. City managers should prioritize service planning and operational control in these areas during the peak period. Practical measures include dynamic service enhancements during peak hours, such as deploying dedicated commuter express routes, increasing service frequency, and implementing other short-term capacity adjustments to accommodate rapidly rising demand and to mitigate crowding spillover across the wider network.

The network topological analysis suggests that the bus system exhibits approximate small-world characteristics that a relatively small subset of stops have high degree centrality, functioning as key interchange hubs that connect many intermediate stops through multiple routes. Such highly connected hubs can be operationally vulnerable: disruptions at these nodes (e.g., due to incidents, congestion, or severe overcrowding) may generate disproportionate network-wide impacts (Du et al. 2020; Ma et al. 2023). Therefore, agencies should particularly prioritize these interchange hubs. Backup strategies, such as pre-planned diversion schemes, short-turn operations, and rapid-response protocols, can help resolve sudden disruptions in a timely manner and reduce the likelihood that failures propagate through the network.

The results further indicate that network topological features provide complementary explanatory value beyond urban land-use characteristics in accounting for temporal passenger demand patterns. CBDs concentrate business and commercial activities and often exhibit pronounced evening peaks, reflecting after-work travel. In these contexts, bus services play an important role in redistributing passengers from city centers to surrounding residential areas. Moreover, stops located in CBDs tend to occupy structurally prominent positions in the network, characterized by higher connectivity and distinctive local clustering compared with stops in other areas. Therefore, city managers should incorporate topological characteristics alongside land use features to enhance the understanding of temporal passenger volume patterns, supporting more effective and efficient urban bus network planning.

6 Conclusion

This study develops a framework to investigate temporal passenger ridership patterns in relation to the network topological characteristics of the urban public bus system. Topological measures provide an informative perspective on the role of bus stops within the network context, complementing widely used built environment and land use attributes. By linking stop-level temporal ridership profiles with topological properties of the transit network, this research extends the application of complex network theory in transport analysis and moves beyond examining physical network structure in isolation. The findings offer new insights into the functioning of public bus networks and the travel behaviors they accommodate. Using the Singapore bus system as a case study, three main conclusions are drawn.

(1) Among several clustering approaches, the K-means method was selected to cluster workday hourly boarding volumes into five typical temporal patterns. Based on peak period dynamics (i.e., morning, midday, and evening), spatial distribution, and surrounding land use composition, these clusters were interpreted as reflecting distinct activity patterns and urban functional areas: Mixed-use CBD (Cluster 1), western job hubs (Cluster 2), peripheral residential regions (Cluster 3), near-central residential regions (Cluster 4), and school-oriented zones (Cluster 5).

(2) The public bus network was represented using both L-space and P-space directed graph models, and seven topological indicators were used to quantify each stop. The L-space representation more strongly exhibits scale-free tendencies, indicating that a small proportion of bus interchange hubs play dominant roles in concentrating and redistributing passenger flows. In addition, the bus network displays small-world properties, suggesting a high level of overall accessibility and structural cohesion.

(3) The MNL model revealed the statistically significant relationship between the ridership patterns and topological features. Estimation results show that Clusters 2 (western job hubs), 3 (peripheral residential regions), and 5 (school-oriented zones) are most strongly differentiated by topology, whereas Cluster 4 is less distinguishable. Incorporating topological features enhances the understanding of temporal bus passenger volume patterns from a network structural perspective by capturing connectivity, transitivity, and transfer potential, thereby complementing widely used land use attributes.

This study has two main limitations. First, topological metrics can be sensitive to edge definitions and weighting schemes, which may influence the inferred associations with ridership patterns. Although this study employed unweighted edges, alternative weight definitions, such as travel distance (Wu and Kim 2020), service frequency (Abdelaty et al. 2020; Luo et al. 2020), and segment-level passenger volume (Sun and Guan 2016), are widely used. Future work will systematically compare these approaches to identify appropriate weighting strategies for bus network representations. Second, this research focused on bus systems, similar ridership profiles can also be observed in other transport modes, including metro (Gan et al. 2020), taxi systems (Choi et al. 2022). Extending the proposed framework to these modes would strengthen the generalizability of the findings and provide a more comprehensive understanding of multimodal urban mobility.

Data Availability Statement

The data that support the findings of this study are openly available from the Singapore Land Transport Authority (LTA) DataMall Platform: https://datamall.lta.gov.sg.

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References

Abdelaty, H., M. Mohamed, M. Ezzeldin, and W. El-Dakhakhni. 2020. "Quantifying and classifying the robustness of bus transit networks." *Transportmetrica A: Transport Science*, 16 (3): 1176–1216. https://doi.org/10.1080/23249935.2020.1720042.

Barabási, A.-L., and M. Pósfai. 2016. *Network science*. Cambridge, United Kingdom: Cambridge University Press.

Calinski, T., and J. Harabasz. 1974. "A dendrite method for cluster analysis." *Communications in Statistics - Theory and Methods*, 3 (1): 1–27. https://doi.org/10.1080/03610927408827101.

Choi, J., W. No, M. Park, and Y. Kim. 2022. "Inferring land use from spatialtemporal taxi ride data." *Applied Geography*, 142: 102688. https://doi.org/10.1016/j.apgeog.2022.102688.

Davies, D. L., and D. W. Bouldin. 1979. "A Cluster Separation Measure." *IEEE Transactions on Pattern Analysis and Machine Intelligence*, PAMI-1 (2): 224–227. https://doi.org/10.1109/TPAMI.1979.4766909.

Dou, M., Y. Wang, and S. Dong. 2021. "Integrating Network Centrality and Node-Place Model to Evaluate and Classify Station Areas in Shanghai." *ISPRS International Journal of Geo-Information*, 10 (6): 414. https://doi.org/10.3390/ijgi10060414.

Du, Z., J. Tang, Y. Qi, Y. Wang, C. Han, and Y. Yang. 2020. "Identifying critical nodes in metro network considering topological potential: A case study in Shenzhen city—China." *Physica A: Statistical Mechanics and its Applications*, 539: 122926. https://doi.org/10.1016/j.physa.2019.122926.

El Mahrsi, M. K., E. Come, L. Oukhellou, and M. Verleysen. 2017. "Clustering Smart Card Data for Urban Mobility Analysis." *IEEE Transactions on Intelligent Transportation Systems*, 18 (3): 712–728. https://doi.org/10.1109/TITS.2016.2600515.

Fagiolo, G. 2007. "Clustering in complex directed networks." *Physical Review E*, 76 (2): 026107. https://doi.org/10.1103/PhysRevE.76.026107.

von Ferber, C., T. Holovatch, Yu. Holovatch, and V. Palchykov. 2009. "Public transport networks: empirical analysis and modeling." *The European Physical Journal B*, 68 (2): 261–275. https://doi.org/10.1140/epjb/e2009-00090-x.

Gan, Z., M. Yang, T. Feng, and H. Timmermans. 2020. "Understanding urban mobility patterns from a spatiotemporal perspective: daily ridership profiles of metro stations." *Transportation*, 47 (1): 315–336. https://doi.org/10.1007/s11116-018-9885-4.

Gecchele, G., R. Ceccato, and M. Gastaldi. 2019. "Road Network Vulnerability Analysis: Case Study Considering Travel Demand and Accessibility Changes." *Journal of Transportation Engineering, Part A: Systems*, 145 (7): 05019004. https://doi.org/10.1061/JTEPBS.0000252.

He, Y., D. Lin, J. Gao, E. He, L. Zhang, C. Yang, L. Sun, S. Ye, and Z. Chen. 2025. "Resilience Assessment of Highway Networks under the Influence of Natural Disasters in the Plateau Mountainous Areas of Western Sichuan, China." *Journal of Transportation Engineering, Part A: Systems*, 151 (10): 04025084. https://doi.org/10.1061/JTEPBS.TEENG-9156.

Huang, A., H. M. Zhang, W. Guan, Y. Yang, and G. Zong. 2015. "Cascading Failures in Weighted Complex Networks of Transit Systems Based on Coupled Map Lattices." *Mathematical Problems in Engineering*, 2015: 1–16. https://doi.org/10.1155/2015/940795.

Katz, L. 1953. "A new status index derived from sociometric analysis." *Psychometrika*, 18 (1): 39–43. https://doi.org/10.1007/BF02289026.

Kim, K. 2018. "Exploring the difference between ridership patterns of subway and taxi: Case study in Seoul." *Journal of Transport Geography*, 66: 213–223. https://doi.org/10.1016/j.jtrangeo.2017.12.003.

Kim, S., and K. Jang. 2022. "Effects of land use on time-of-day transit ridership patterns." *Transportmetrica A: Transport Science*, 18 (3): 1777–1793. https://doi.org/10.1080/23249935.2021.1970046.

Li, X., G. Xing, X. Qian, Y. Guo, W. Wang, and C. Cheng. 2022. "Subway Station Accessibility and Its Impacts on the Spatial and Temporal Variations of Its Outbound Ridership." *Journal of Transportation Engineering, Part A: Systems*, 148 (12): 04022106. https://doi.org/10.1061/JTEPBS.0000766.

Lin, J., and Y. Ban. 2013. "Complex Network Topology of Transportation Systems." *Transport Reviews*, 33 (6): 658–685. https://doi.org/10.1080/01441647.2013.848955.

Liu, S., E. Yao, and B. Li. 2019. "Exploring urban rail transit station-level ridership growth with network expansion." *Transportation Research Part D: Transport and Environment*, 73: 391–402. https://doi.org/10.1016/j.trd.2018.04.006.

Liu, X., L. Gong, Y. Gong, and Y. Liu. 2015. "Revealing travel patterns and city structure with taxi trip data." *Journal of Transport Geography*, 43: 78–90. https://doi.org/10.1016/j.jtrangeo.2015.01.016.

Lordan, O., J. M. Sallan, P. Simo, and D. Gonzalez-Prieto. 2014. "Robustness of the air transport network." *Transportation Research Part E: Logistics and Transportation Review*, 68: 155–163. https://doi.org/10.1016/j.tre.2014.05.011.

Luo, D., O. Cats, and H. van Lint. 2020. "Can passenger flow distribution be estimated solely based on network properties in public transport systems?" *Transportation*, 47 (6): 2757–2776. https://doi.org/10.1007/s11116-019-09990-w.

Ma, Z., J. Liu, S. I.-J. Chien, X. Hu, and Y. Shao. 2023. "Identifying Critical Stations Affecting Vulnerability of a Metro Network Considering Passenger Flow and Cascading Failure: Case of Xi'an Metro in China." *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, 9 (2): 04023014. https://doi.org/10.1061/AJRUA6.RUENG-1013.

Meng, Y., X. Tian, Z. Li, W. Zhou, Z. Zhou, and M. Zhong. 2020. "Comparison analysis on complex topological network models of urban rail transit: A case study of Shenzhen Metro in China." *Physica A: Statistical Mechanics and its Applications*, 559: 125031. https://doi.org/10.1016/j.physa.2020.125031.

Newman, M. E. J. 2010. *Networks: An Introduction*. Oxford ; New York: Oxford University Press.

Page, L., S. Brin, R. Motwani, and T. Winograd. 1999. *The PageRank Citation Ranking: Bringing Order to the Web.* Technical Report. Stanford InfoLab.

Pei, T., S. Sobolevsky, C. Ratti, S.-L. Shaw, T. Li, and C. Zhou. 2014. "A new insight into land use classification based on aggregated mobile phone data." *International Journal of Geographical Information Science*, 28 (9): 1988–2007. https://doi.org/10.1080/13658816.2014.913794.

de Regt, R., C. von Ferber, Y. Holovatch, and M. Lebovka. 2019. "Public transportation in Great Britain viewed as a complex network." *Transportmetrica A: Transport Science*, 15 (2): 722–748. https://doi.org/10.1080/23249935.2018.1529837.

Shanmukhappa, T., I. Wang-Hei Ho, C. K. Tse, and K. K. Leung. 2019. "Recent Development in Public Transport Network Analysis From the Complex Network Perspective." *IEEE Circuits and Systems Magazine*, 19 (4): 39–65. https://doi.org/10.1109/MCAS.2019.2945211.

Sienkiewicz, J., and J. A. Hołyst. 2005. "Statistical analysis of 22 public transport networks in Poland." *Physical Review E*, 72 (4): 046127. https://doi.org/10.1103/PhysRevE.72.046127.

Singapore Department of Statistics. 2021. *Singapore Census of Population 2020, Statistical Release 2: Households, Geographic Distribution, Transport and Difficulty in Basic Activities*.

Sun, D. (Jian), and S. Guan. 2016. "Measuring vulnerability of urban metro network from line operation perspective." *Transportation Research Part A: Policy and Practice*, 94: 348–359. https://doi.org/10.1016/j.tra.2016.09.024.

Sun, D., Y. Zhao, and Q.-C. Lu. 2015. "Vulnerability Analysis of Urban Rail Transit Networks: A Case Study of Shanghai, China." *Sustainability*, 7 (6): 6919–6936. https://doi.org/10.3390/su7066919.

Taylor, M. A. P. 2017. *Vulnerability analysis for transportation networks*. Amsterdam, Netherlands ; Cambridge, MA: Elsevier.

Tocchi, D., C. Sys, A. Papola, F. Tinessa, F. Simonelli, and V. Marzano. 2022. "Hypergraph-based centrality metrics for maritime container service networks: A worldwide application." *Journal of Transport Geography*, 98: 103225. https://doi.org/10.1016/j.jtrangeo.2021.103225.

Wang, J., N. Zhang, H. Peng, Y. Huang, and Y. Zhang. 2022. "Spatiotemporal Heterogeneity Analysis of Influence Factor on Urban Rail Transit Station Ridership." *Journal of Transportation Engineering, Part A: Systems*, 148 (2): 04021115. https://doi.org/10.1061/JTEPBS.0000639.

Wang, X., Y. Li, R. Zhang, J. Liu, and L. Gao. 2024. "Comprehensive Network-Level Urban Road Asset Valuation Method Integrating Physical and Social Values." *Journal of Transportation Engineering, Part A: Systems*, 150 (7): 04024024. https://doi.org/10.1061/JTEPBS.TEENG-8244.

Ward, J. H. 1963. "Hierarchical Grouping to Optimize an Objective Function." *Journal of the American Statistical Association*, 58 (301): 236–244. https://doi.org/10.1080/01621459.1963.10500845.

Wu, C., and I. Kim. 2020. "Analyzing the structural properties of bike-sharing networks: Evidence from the United States, Canada, and China." *Transportation Research Part A: Policy and Practice*, 140: 52–71. https://doi.org/10.1016/j.tra.2020.07.018.

Zhao, X., Y. Wu, G. Ren, K. Ji, and W. Qian. 2019. "Clustering Analysis of Ridership Patterns at Subway Stations: A Case in Nanjing, China." *Journal of Urban Planning and Development*, 145 (2): 04019005. https://doi.org/10.1061/(ASCE)UP.1943-5444.0000501.